

The Power of Noise: Redefining Retrieval for RAG Systems

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Paper Link - <https://arxiv.org/pdf/2401.14887>

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Overview of the Problem

Retrieval Augmented Generation combines a retriever that finds relevant information from large collection of documents and a generator which is large language model that uses those documents to produce answers. The Goal is to improve accuracy and reduce hallucinations by providing information to the model in real time.

However, the paper finds that even though the retriever returns top-ranked documents the LLMs performance can decline. Instead of helping the LLM, these documents are distracting LLM which causes irrelevant responses. and sometimes adding unrelated documents improves LLMs accuracy. This unexpected behaviour shows that the retriever in RAG systems needs to be reevaluated, especially as modern AI search now focuses on context, user intent, and personalization, not just keyword matching.

Literature and Proposed Solution

Earlier research on Retrieval Augmented Generation focused mainly on improving retrieval accuracy by ensuring that the top k retrieved documents were the most relevant for LLM. Methods like BM25 a sparse retriever based on keyword matching and Contriever a dense retriever using neural embeddings were widely used.

However this paper challenges these assumptions, The authors argues that retrieval quality cannot be judged only by traditional IR metrics when the goal is to support a generative model. Even top k retrieved documents can contain distracting information that causes LLM to produce wrong outputs. Author tested how RAG systems behave when they added relevant, distracting or random documents, change the number of retrieved documents and adjusted the documents appearing in the prompt. Based on the experiment the author proposed a new strategy by only using few very important documents, add some random documents to increase diversity and prevent confusion and place the most relevant documents closest to the query in the prompt. This approach redefined the role of the retriever from by simply ranking documents to balancing context which helped LLM reason more effectively.

Results and Limitations

The experiments were conducted on multiple LLM models which includes Llama-2, MPT, Phi-2 using datasets Natural Questions (NQ). Both dense retrievers like Contriever and sparse retrievers like BM25 were tested. The Results -

- Distracting Documents reduce accuracy – High Scoring but irrelevant documents can mislead the model and decrease the answers accuracy
- Random Documents can improve accuracy – Adding noisy documents can often improve results. The performance can improve upto 35% as random content can prevent model to over focus on wrong details.
- Few documents worked better – Using 3-5 relevant documents was more effective than many.
- Position Matters – Placing the most relevant documents near the question in the prompt improved quality.
- Works on both retrieval types – Both dense and sparse retrieval methods showed similar trends.

Over all the study proved that interaction between the retriever and the LLM is complex.

- Lack of Full Explanation – Random noise improves performance is not fully understood. Its an hypothesis that because of attention diversity we are getting better results.
- Model Specific Behaviour – This effect may vary between Models.
- Dataset Scope – Most experiments were on Wikipedia based dataset (NQ dataset) so results might be different for specialised or domain specific content.
- Evaluation Method – Evaluation only matches accuracy with exact answer, not partially correct responses.

Summary

This paper shows that retrieval in RAG systems needs to evolve just like modern AI search engines that now use semantics, intent, and personalization. Instead of relying only on top-ranked documents, we should balance relevant and random information and organize it thoughtfully in the prompt. This approach helps LLMs reason better, stay grounded, and provide more accurate and meaningful answers. the paper proves that sometimes, a little bit of noise can actually make AI smarter.